AI Cylinder



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Artificial insemination (AI) cylinder

What it is used for

An AI cylinder stores liquid nitrogen for preserving semen straws or other approved biological material at cryogenic temperatures. Correct handling, level monitoring, and secure storage help maintain sample viability while reducing cold-burn and oxygen-displacement risks.

Main parts

- Vacuum-insulated cylinder reduces heat transfer and maintains cryogenic conditions.
- Neck opening provides access to the storage chamber.
- Canisters or racks hold labelled straws or approved sample containers.
- Neck plug or cover limits unnecessary nitrogen loss while allowing safe venting as designed.
- Liquid nitrogen provides the cryogenic cooling medium.
- Level measuring device or monitoring method checks the nitrogen level where provided.
- Identification labels and inventory records track stored samples.
- Protective outer casing supports safe handling and transport.

Before you start

1. Only trained and authorised personnel should handle the cylinder and biological samples.
2. Read the manufacturer's instructions and the facility's cryogenic material handling procedure.
3. Inspect the cylinder, neck, lid, handles, labels, and base for damage or abnormal frost patterns.
4. Check the liquid nitrogen level using the approved measuring method without leaning over the opening.
5. Ensure the room is adequately ventilated and an oxygen-deficiency monitor is available where required.
6. Wear cryogenic gloves, eye protection or a face shield, closed footwear, and suitable protective clothing.

How to use

1. Place the cylinder upright on a stable, level surface in the designated ventilated area.
2. Confirm the sample inventory and identify the required canister or rack before opening the cylinder.
3. Remove the cover carefully and allow the neck area to remain clear of obstructions.
4. Use the approved lifting tool to raise the correct canister without reaching into the neck opening.
5. Locate the labelled sample using the inventory and minimise the time the cylinder remains open.
6. Return the canister to its correct position and ensure stored samples remain securely arranged.

7. Close the cylinder with the specified cover or neck plug without sealing it in a way that prevents designed venting.
8. Measure and record the liquid nitrogen level according to the facility schedule.
9. Refill only according to the approved procedure using compatible transfer equipment and trained personnel.
10. Secure the cylinder upright, update sample records, and report any damage, abnormal frost, or low nitrogen level.

⚠ Safety

- Liquid nitrogen can cause severe cold burns and can displace oxygen in poorly ventilated spaces.
- Never seal the cylinder airtight or block its designed venting path.
- Never look directly into the opening or place the face over the neck.
- Use only cryogenic-rated gloves and face or eye protection; ordinary thin gloves are not sufficient.
- Do not handle the cylinder alone during high-risk transfers or move it by dragging or tipping.
- If an oxygen-deficiency alarm sounds or breathing difficulty occurs, leave the area immediately and follow the emergency procedure.

Care & storage

- Keep the cylinder upright, clean, and protected from impact.
- Monitor nitrogen level at the required frequency and maintain sample inventory records.
- Inspect the neck, cover, handles, and exterior regularly for damage or unusual frost.
- Arrange authorised inspection if vacuum insulation may have failed or nitrogen consumption increases unexpectedly.

Common mistakes to avoid

- Allowing the cylinder to remain open longer than necessary.
- Blocking the vent or sealing the cylinder airtight.
- Handling liquid nitrogen without appropriate cryogenic PPE or ventilation.
- Failing to monitor nitrogen level and sample location records.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.